BD-Statistics (M/M/1)

for $\rho < 1$ (stable)

$$\pi_i = P_r(Q=i) = (1-\rho)\rho^i$$

Derive:

Idle server $\quad \pi_0 = (1-\rho)\rho^0 = 1-\rho$

Avg Q Len $\quad \mathbb{E}\{Q\} = \dfrac{\rho}{1-\rho}$ $\rightarrow$ (Util./Frac time Busy) $\rightarrow$ (idle Frac time)

if $\lambda = 4 \quad \mu = 6 \quad \dfrac{\rho}{1-\rho} = \dfrac{0{,}67}{0{,}33} = \mathbb{E}\{Q\} = 2$

Avg delay $\quad \mathbb{E}\{S\} = \dfrac{\mathbb{E}\{Q\}}{\lambda} = \dfrac{1}{\mu - \lambda}$

Buffer overflow (PASTA)

$$P_r(Q^{(a)} \geq B) = P_r(Q \geq B) = \rho^B$$

Note: Increase server speed always better

M/M/2 $\rightarrow$